

SIHYEON KIM

Master's Student in City and Regional Planning
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PROFESSIONAL SUMMARY

Urban planning and architecture researcher specializing in urban morphology, microclimate simulation, and heat mitigation strategies. Research focuses on the interaction between urban form typology and Urban Heat Island (UHI) dynamics, integrating AI-based planning tools and cost-effectiveness analysis. Experienced in interdisciplinary environments spanning architecture, urban planning, and data-driven climate analysis, with international academic exposure and multiple peer-reviewed conference presentations.

EDUCATION

Seoul National University M.S. in City and Regional Planning (City & Energy Planning Track) Cumulative GPA: 4.30 / 4.30	Seoul, Korea 2025 – Present
Soongsil University Bachelor of Architecture (B.Arch), 5-year program Major GPA: 4.10 / 4.50 Cumulative GPA: 4.15 / 4.50 Departmental Honors Award (3rd rank overall)	Seoul, Korea 2025

INTERNATIONAL EDUCATION

Chiles High School AP Art Awards; City Hall Exhibitions	Florida, USA 2015.08 – 2016.08
Deer Lake Middle School Graduated	Florida, USA 2014.08 – 2015.08
Meeker Elementary School Kindergarten – Grade 1	Iowa, USA 2006.01 – 2006.12

RESEARCH EXPERIENCE

Research Assistant, City Energy Lab <i>Graduate School of Environmental Studies, Seoul National University Supervisor: Prof. Steven Jige Quan</i> - Conduct research on urban heat mitigation strategies and morphological pattern analysis in Seoul's urban regeneration projects - Apply microclimate simulation (SOLWEIG, ENVI-met) and spatial optimization methods (genetic algorithms)	2024.10 – Present
UN-Habitat Global South Urban Shrinkage Project Participated in international research collaboration on urban shrinkage in the Global South	2025.07
Seoul Urban Regeneration Project, City Energy Lab Contributed to research on energy and climate performance of urban regeneration areas under Prof. Jige Quan	2024.01 –

PUBLICATIONS

Journal Articles (Under Review / In Preparation) Quan, J., Kim, S. (2025). Revealing Hidden Archetypes of Urban Morphological Patterns in Seoul. Manuscript in preparation. (Submission accepted) Kim, S., Quan, J. (2026). Evaluation of Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. In preparation. Kim, S., Quan, J. (2026). Budget-Constrained Heat Mitigation Spatial Optimization in Three Seoul Urban Regeneration Areas: Using SOLWEIG and Genetic Algorithms. In preparation.

CONFERENCE PRESENTATION

Kim, S. (2026). Evaluation Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 2026 ACSP 66th Annual Conference, United States, October 10–12.
Kim, S. (2026). Cost-Effective Heat Mitigation across Urban Regeneration Typologies: Implications for Urban Microclimate in Seoul. 20th International Association for China Planning Conference (IACP), Xi'an, China, July 10–12.
Kim, S. (2026). Thermal Performance and Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. The 2nd Workshop on AI for Urban Planning, 40th Annual AAAI Conference on Artificial Intelligence, Singapore, January 26.

Kim, S. (2026). Revealing Hidden Archetypes of Urban Morphological Patterns in Seoul. The 2nd Workshop on AI for Urban Planning, 40th Annual AAAI Conference on Artificial Intelligence, Singapore, January 26.

Kim, S. (2025). Assessing Urban Heat Mitigation Strategies in Seoul’s Regeneration Projects. 19th International Association for China Planning Conference (IACP), Xiamen, China, July 3–7.

Kim, S., Quan, J. (2025). Evaluating the Effectiveness of Heat Mitigation Strategies in Seoul’s Urban Regeneration Projects. Urban Design Institute of Korea (UDIK) Spring Conference, Chuncheon, Korea, April 18–19. Proceedings, pp. 792–795.

HONORS & AWARDS

ACADEMIC & RESEARCH

- Best Cross-disciplinary Integration Award, Participatory Urban Regeneration Empowered by Artificial Intelligence — The 2nd AI4CITIES International Conference & Inaugural AI4CITIES Summer Workshop, Tongji University, Shanghai, China (July 2025)

ARCHITECTURE DESIGN

- Departmental Honors Award (3rd place graduation), Soongsil University (2025)
- Graduation Design Award (4th place), Soongsil University (2024)
- Encouragement Award, Modular Urban Regeneration Project — Korea Space Design Association National Competition (2023)
- Award Winner, Eco-friendly Office Building Design — Korean Society of Basic Design & Art; Exhibited at Sejong Center for the Performing Arts (2023)
- Outstanding Architectural Design Award — Soongsil University Design Award (1st place) (2021)

FINE ARTS

- Group Exhibition, The 26th Korean Painting Status Exhibition (March 2026)
- 한국여성회화대전 (One Drop City series, Oil Pastel & Acrylic) (March 2026)
- Award-winning work exhibited at Korea Art Museum (4 Planet), JoongAng Ilbo Central Painting Competition (2024)
- Award Winner (Lucid Dream), Korean Society of Basic Design & Art, Oil Painting Division — Sejong Center Exhibition (2023)
- Award Winner (Minerva of the City), JoongAng Ilbo Central Painting Competition (2023)
- Award Winner (Lucid Dream), JoongAng Ilbo Central Painting Competition (2022)
- 1st Prize, AP Art Awards, Chiles High school, Tallahassee, Florida (2014–2015)
- City Hall Winter Youth Art Exhibition, Deerlake middle school, Tallahassee, Florida (2014–2015)

SCHOLARSHIPS & TEACHING

- Teaching Assistant, City and Regional Planning, Graduate School of Environmental Studies, Seoul National University (2026)
- Full Scholarship, Architecture Design (5-year program), Soongsil University (2019–2025)

PROFESSIONAL EXPERIENCE

Architecture Intern, Design Somnium

GCC Multi-purpose Cultural Center Remodeling Project — On-site construction and design coordination

2024.01 – 2024.07

SKILLS

Architectural Design & Building Simulation: AutoCAD, SketchUp, Rhino 3D, V-Ray, IES VE, DesignBuilder

Urban Analytics & Environmental Simulation: ENVI-met, SOLWEIG, Grasshopper, Ladybug Tools, ArcGIS / GIS, Python

Visual Communication & Design: Adobe Photoshop, Illustrator, InDesign, Figma, Procreate, Acrylic Painting

CERTIFICATIONS & LANGUAGES

- ESG SNU Youth Forum Certification (2026)
- Rhino Certification (2025.08)
- OPIC: IH | TEPS: 403
- Languages: Korean (Native), English (Professional working proficiency)

VOLUNTEER EXPERIENCE

- Sign Language Interpreter Volunteer — Real-time interpretation for church services (2019 – Present)
- Volunteer Mentor, Cambodia — Educational talent donation program for elementary school students (August 2019)